

Deep Learning for Multi-Class Skin Lesion Classification on HAM10000: A Literature Survey and Base-Paper Selection

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Abstract—Skin cancer is among the most commonly diagnosed cancers worldwide, and melanoma accounts for a disproportionate share of its mortality despite being curable when detected early. Automated analysis of dermatoscopic images has therefore become an active computer-vision problem, and the HAM10000 dataset has become its most widely used public benchmark. This paper presents Milestone 1 of a research-reproduction project: a survey of ten representative works on dermatoscopic lesion classification, a comparative analysis of their methods, datasets and reported metrics, and the selection of a single base paper for reproduction. We organise the literature into five themes: clinical benchmarks, dataset and challenge protocols, transfer-learning baselines, attention and class-imbalance methods, and data-quality auditing. From this survey we select Datta *et al.*'s Soft-Attention model as the base paper, because it is evaluated on HAM10000, releases complete public code for *both* its baseline and its proposed method, and reports precision, sensitivity and AUC rather than accuracy alone. We conclude with a filled reproducibility checklist that identifies the specific gaps we expect to encounter, and a reproduction protocol for Milestone 2. No clinical claims are made in this work.

Index Terms—skin lesion classification, melanoma, HAM10000, dermoscopy, convolutional neural networks, transfer learning, attention, reproducibility, class imbalance

I. INTRODUCTION

Skin cancer is the most commonly diagnosed cancer in several populations [1]. Melanoma represents only a small fraction of skin-cancer cases but causes the majority of skin-cancer deaths, and its prognosis depends strongly on the stage at which it is detected. Dermoscopy, a non-invasive magnified imaging technique, improves diagnostic accuracy over naked-eye examination, but its interpretation is subjective and requires substantial training.

This has motivated a decade of work on automated dermatoscopic image analysis. The field changed direction in 2017, when Esteva *et al.* showed that a single convolutional neural network (CNN) trained on a large clinical corpus could match board-certified dermatologists on binary classification [1]. Progress after that point was limited less by architectures than by data: publicly available dermatoscopic datasets were small, single-source and demographically narrow. The release of the HAM10000 dataset in 2018 [2] directly addressed this gap and

has since become the default public benchmark for multi-class lesion classification.

A. Aim of this Milestone

The goal of this project is not to propose a novel architecture, but to *reproduce* a published result and then propose, implement and defend one controlled change of our own. Milestone 1 therefore has three deliverables:

- 1) a survey of recent literature on the chosen topic, covering methods, datasets and evaluation metrics;
- 2) the selection of one base paper, preferably with public code and a downloadable dataset; and
- 3) a completed reproducibility checklist for that paper.

B. Contributions of this Report

- We survey ten representative papers and organise them into a five-theme taxonomy (Section III).
- We tabulate their methods, datasets, reported metrics and code availability for direct comparison (Table II).
- We define explicit selection criteria and justify our choice of base paper against them (Section IV).
- We complete a reproducibility checklist that names the concrete gaps we expect to hit during reproduction (Table III).
- We identify a documented evaluation-protocol weakness in the base paper that will motivate our Milestone 3 hypothesis.

II. PROBLEM DEFINITION AND DATASET

A. Task

We address *seven-class classification of dermatoscopic images*. Given a single RGB dermatoscopic image x , the model must predict a label $y \in \mathcal{Y}$, where $\mathcal{Y}$ contains the seven diagnostic categories of HAM10000. This is a single-label, multi-class problem; it is not segmentation and it is not a binary melanoma-versus-benign task, although much of the surveyed literature addresses the binary formulation.

B. The HAM10000 Dataset

HAM10000 (“Human Against Machine with 10000 training images”) was assembled by Tschandl *et al.* from two sites, a dermatology practice in Australia and the Medical University of Vienna [2]. It contains 10,015 dermatoscopic images at 600×450 pixels across seven classes. More than half of the lesions are confirmed by histopathology; the remainder are confirmed by follow-up, expert consensus or confocal microscopy. The dataset is distributed through the Harvard Dataverse and is mirrored on Kaggle, satisfying the “downloadable dataset” requirement of this project.

Table I gives the class distribution. Two properties dominate every design decision in the surveyed literature.

TABLE I
CLASS DISTRIBUTION OF THE HAM10000 DATASET [2].

Code	Diagnosis	Images	%
nv	Melanocytic nevi	6,705	66.9
mel	Melanoma	1,113	11.1
bkl	Benign keratosis	1,099	11.0
bcc	Basal cell carcinoma	514	5.1
akiec	Actinic keratoses	327	3.3
vasc	Vascular lesions	142	1.4
df	Dermatofibroma	115	1.1
Total		10,015	100.0

(i) **Severe class imbalance.** The largest class is roughly 58 times larger than the smallest. A trivial classifier that always predicts `nv` achieves 66.9% accuracy while detecting no melanoma at all. Accuracy is therefore an actively misleading metric on this dataset, and we report sensitivity (recall), precision and AUC throughout, as required by the project brief.

(ii) **Repeated lesions.** The 10,015 images correspond to only 7,470 distinct lesions; 2,545 images are additional views of a lesion that already appears elsewhere in the dataset. The metadata exposes this through the `lesion_id` field. If a dataset split is performed at the image level rather than the lesion level, near-duplicate views of the same physical lesion can land in both the training and test partitions, inflating reported performance. As Section III-E shows, this is a documented and widespread problem in this literature, and it is central to our choice of base paper.

C. Evaluation Metrics

Following the ISIC challenge protocol [3] and the constraints of this project, we adopt: per-class and macro-averaged *sensitivity/recall*, *precision*, *macro F1*, and *AUC*. Overall accuracy is reported only for comparability with prior work and is never used as the primary criterion.

III. LITERATURE SURVEY

We surveyed ten core papers, grouped below by the role they play in the field. Table II summarises them.

A. Clinical Benchmarks: What Counts as “Good”

Esteva *et al.* (2017) [1] trained a single Inception-v3 network on 129,450 clinical images spanning 2,032 diseases and matched 21 board-certified dermatologists on biopsy-proven binary tasks (keratinocyte carcinoma vs. benign seborrheic keratosis, and malignant melanoma vs. benign nevus). Its importance here is methodological rather than architectural: it established that transfer learning from ImageNet, applied to a large enough dermatological corpus, is sufficient for clinically comparable performance. It does not use HAM10000.

Haenssle *et al.* (2018) [4] compared a CNN against 58 dermatologists from 17 countries on dermoscopic melanoma recognition. The dermatologists achieved a mean sensitivity of 86.6% and mean specificity of 71.3%; the CNN achieved higher specificity at equivalent sensitivity. This paper supplies the human operating point against which our own sensitivity numbers should be contextualised, and it is the reason we report sensitivity as a primary metric.

Tschandl *et al.* (2020) [7] evaluated human–computer collaboration rather than replacement, and found that AI-assisted clinicians outperformed either the AI or the clinician alone, with the least experienced readers gaining the most. This frames the honest deployment claim for any work in this area and directly informs our limitations section.

B. Datasets and Challenge Protocols

Tschandl *et al.* (2018) [2] is the HAM10000 data descriptor and is a mandatory citation for this project. Beyond the class counts, it documents the acquisition pipeline, the multi-source composition, and critically the presence of multiple images per lesion — the property that Section III-E shows most subsequent papers ignore.

Codella *et al.* (2018) [3] describes the ISIC 2017 challenge hosted at ISBI, which standardised the evaluation protocol for this field: fixed train/validation/test partitions, and balanced multi-class accuracy rather than raw accuracy. It is worth noting that the ISIC 2018 Task 3 training set is HAM10000, so a number of papers that describe themselves as “ISIC 2018” results are in fact reporting on our dataset.

C. Transfer-Learning Baselines on HAM10000

Chaturvedi *et al.* (2020) [6] fine-tuned a pretrained MobileNet on HAM10000 for the full seven-class task, reporting 83.1% categorical accuracy, 91.36% top-2 accuracy and weighted precision/recall/F1 of 0.89/0.83/0.83. It is the cheapest credible baseline in the literature and remains a useful lower bound.

Ali *et al.* (2021) [9] fine-tuned the full EfficientNet family, B0 through B7, on HAM10000 under a single consistent protocol. EfficientNet-B4 performed best with 87.91% top-1 accuracy and an F1 of 0.87. The more interesting result is that the largest variants (B6, B7) did *not* outperform the mid-sized ones, indicating that on a dataset of this size, capacity is not the binding constraint — data quantity and imbalance are. This is a well-designed capacity ablation and a strong

secondary candidate for reproduction, but the authors do not release official code.

D. Attention, Imbalance and Augmentation

Datta et al. (2021) [8] introduced a *Soft-Attention* module that can be inserted into an existing CNN backbone to amplify informative feature-map regions and suppress noise. They evaluated five backbones (VGG16, ResNet34, ResNet50, Inception-ResNet-v2, DenseNet201) with and without the module. The best configuration, Inception-ResNet-v2 with Soft-Attention, reached 93.7% precision and 98.4% AUC on HAM10000, improving precision over the corresponding baseline by 3.2%. On ISIC-2017 it reached 91.6% sensitivity. Crucially, the authors release complete public code containing *both* the with-attention and without-attention variants, plus a Grad-CAM notebook for qualitative analysis. This is the paper we select; see Section IV.

Yao et al. (2022) [10] attacked the imbalance problem directly, combining a Modified RandAugment policy, DropBlock regularisation, a class-balanced loss and multi-weighted new loss. They showed that a single well-regularised model of *moderate* capacity can outperform large ensembles on small, imbalanced dermatological datasets — a result that converges with Ali et al.’s capacity finding. The accompanying PyTorch framework is the most professionally engineered public code-base we found in this area, and its ISIC-2018 configuration corresponds to HAM10000.

Perez et al. (2018) [5] systematically ablated thirteen data-augmentation scenarios across three CNN architectures, including unusual options such as elastic transforms, random erasing and lesion mixing. Their best scenario reached an AUC of 0.882 without external data, exceeding the top ISIC 2017 challenge submission (0.874). We treat this paper primarily as our *methodological template* for Milestone 4: it demonstrates what a rigorous, one-variable-at-a-time ablation study looks like in this exact problem domain.

E. Data Quality and Evaluation Integrity

Three additional works, beyond the core ten, motivate the critical stance we take towards published HAM10000 numbers.

Abhishek et al. (2025) [12] audited HAM10000 and its derivative DermaMNIST and documented duplicate images, leakage of the same lesion across training and test partitions, mislabelled samples and the absence of a well-defined test split. They released corrected (DermaMNIST-C) and extended (DermaMNIST-E) versions along with their analysis code. This paper is the single most useful reference for designing a defensible experimental protocol on this dataset, and it is the evidence base for our Milestone 3 hypothesis.

The practical consequence is visible in the literature. **Shetty et al. (2022)** [13] report 95.18% accuracy on HAM10000 with a custom CNN, against 87.32% for a random-forest baseline. On inspection, however, this result is obtained on a balanced subset of 100 original images per class, expanded by horizontal flipping to 200 per class, and then split 80/20 – a test set of

280 images, roughly 2.8% of HAM10000. Because the flip augmentation is applied *before* the split, a mirrored copy of a training image can appear in the test set. The headline number is therefore not comparable to results computed on the full dataset. We include this paper deliberately: it illustrates that a high reported accuracy is not by itself evidence of a strong method, and it demonstrates why the reproducibility checklist in Section V is necessary.

Finally, **Rehman et al. (2022)** [14] apply explainable-AI techniques to skin-lesion classification, and **Gessert et al. (2020)** [11] won both tasks of the ISIC 2019 challenge with an ensemble of multi-resolution EfficientNets that also incorporates patient metadata. The latter defines the practical performance ceiling of this problem class, though its ensemble scale places it beyond the compute budget of this project. The systematic review of Naseri and Safaei (2025) [15], covering 34 studies published between 2016 and 2024, confirms that CNN-based transfer learning remains the dominant paradigm.

F. A Note on Research Integrity

During this survey we identified a frequently cited paper, “DenseNet-II: an improved deep convolutional neural network for melanoma cancer detection” (*Soft Computing*, 2022), which reported 95.7% accuracy on HAM10000 and which was **retracted by the publisher** [16]. The retraction note states that an investigation found compromised editorial handling and peer review and irrelevant references, and that the publisher no longer has confidence in the results and conclusions. We record this explicitly because the paper still appears in comparison tables in the recent literature. It is excluded from all comparisons in this report.

IV. BASE PAPER SELECTION

A. Selection Criteria

We scored candidates against six criteria derived from the project brief:

- C1. **Dataset match** — evaluated on HAM10000, which must be publicly downloadable.
- C2. **Public code** — an official implementation exists.
- C3. **Baseline included** — the code releases the comparison baseline as well as the proposed method, so that Milestone 2 can verify a reproduction rather than merely obtain a number.
- C4. **Appropriate metrics** — reports sensitivity/recall and AUC, not accuracy alone.
- C5. **Feasible compute** — trainable on a single consumer or free-tier cloud GPU.
- C6. **Extension surface** — offers a clear axis along which one controlled change can be proposed and ablated in Milestones 3–4.

B. Selected Base Paper

S. K. Datta, M. A. Shaikh, S. N. Srihari, and M. Gao, “Soft-Attention Improves Skin Cancer Classification Performance,” *MICCAI Workshop on Interpretability of Machine Intelligence in Medical*

TABLE II
COMPARATIVE SUMMARY OF THE TEN SURVEYED PAPERS. “REPORTED METRIC” USES EACH PAPER’S OWN HEADLINE FIGURE; THESE ARE *not* DIRECTLY COMPARABLE ACROSS ROWS BECAUSE SPLITS, CLASS COUNTS AND PROTOCOLS DIFFER.

#	Year	Paper / Method	Dataset	Reported metric	HAM10000	Code
1	2017	Esteva <i>et al.</i> — Inception-v3 [1]	Private (129k)	Dermatologist-level, binary tasks	No	No
2	2018	Tschandl <i>et al.</i> — HAM10000 descriptor [2]	HAM10000	Dataset paper (10,015 images, 7 classes)	Yes	Yes
3	2018	Codella <i>et al.</i> — ISIC 2017 challenge [3]	ISIC 2017	Challenge protocol & benchmark	No	Yes
4	2018	Haenssle <i>et al.</i> — CNN vs. 58 dermatologists [4]	Private / ISIC	Sens. 86.6%, Spec. 71.3% (human)	No	No
5	2018	Perez <i>et al.</i> — augmentation ablation [5]	ISIC 2017	AUC 0.882 (13 scenarios ablated)	No	Yes
6	2020	Chaturvedi <i>et al.</i> — MobileNet [6]	HAM10000	Acc. 83.1%, top-2 91.4%, F1 0.83	Yes	Partial
7	2020	Tschandl <i>et al.</i> — human–AI collaboration [7]	Multi-source	AI-assisted > AI alone > clinician	Partial	Yes
8	2021	Datta <i>et al.</i> — Soft-Attention [8]	HAM10000	Prec. 93.7%, AUC 98.4%	Yes	Yes
9	2021	Ali <i>et al.</i> — EfficientNet B0–B7 [9]	HAM10000	Acc. 87.91%, F1 0.87 (B4 best)	Yes	No
10	2022	Yao <i>et al.</i> — imbalance-aware single model [10]	ISIC 2017/18/19	Single model > ensembles	Via ISIC-18	Yes

Image Computing (iMIMIC), LNCS vol. 12929, Springer, 2021.

This paper satisfies all six criteria, and it is the only surveyed candidate that satisfies C3. Its public repository is organised as follows:

- Model without Soft Attention/ — five backbone baselines (VGG16, ResNet34, ResNet50, Inception-ResNet-v2, DenseNet201);
- Model With Soft Attention/ — the same five backbones with the proposed module;
- Model with Grad-cam/ — a Grad-CAM notebook, which supports the qualitative failure analysis required at Milestone 5;
- Best Model with different Test Train splits/ — the best model rerun at 15%, 20% and 30% test fractions.

The separation of baseline and method is what makes this paper unusually well-suited to a reproduction project: Milestone 2 can independently reproduce the baseline, then reproduce the claimed improvement, and report both. The paper also reports precision, sensitivity and AUC (C4), the built-in backbone $\times$ attention grid provides a ready-made ablation structure (C6), and Inception-ResNet-v2 at 299×299 on roughly 10k images is trainable on a single free-tier T4 GPU (C5).

C. Why Not the Alternatives

Ali *et al.* [9] is an excellent and clearly specified study, but has no official code, so a “reproduction” would in fact

be a reimplementaion. It is retained as our fallback. Yao *et al.* [10] has the strongest codebase but targets ISIC configurations and a heavier training schedule than our compute budget allows. Gessert *et al.* [11] is an ensemble of multi-resolution networks and is out of budget by a wide margin. Shetty *et al.* [13] reports the highest headline accuracy of any HAM10000 paper we surveyed, but as shown in Section III-E that figure is computed on a 280-image test set built with pre-split augmentation, and is therefore not a reproducible target.

V. REPRODUCIBILITY CHECKLIST

Table III records our assessment of the base paper against a standard reproducibility checklist. We assess it at 4.5/7: genuinely reproducible, with real and clearly identifiable gaps.

A. Verified Detail of the Evaluation Protocol

Because the reproducibility of the headline number depends on it, we inspected the data-preparation code of the base paper directly rather than relying on the text. The split proceeds as follows: the metadata is grouped by `lesion_id`; only lesions contributing exactly one image are retained as the test candidate pool; a stratified 15% sample of that pool forms the test set (828 images), and all remaining images (9,187) form the training set, which is then augmented to 8,000 images per class.

This protocol is *better* than the naive image-level split used by much of the literature: because the test set is drawn only from single-image lesions, no lesion appears on both sides of the split, and augmentation is correctly applied after splitting.

TABLE III
REPRODUCIBILITY CHECKLIST FOR THE SELECTED BASE PAPER [8].

Item	Status	Notes
Source code	Yes	Public GitHub repository; Jupyter notebooks for all five backbones, with and without attention.
Dataset available	Yes	HAM10000, Harvard Dataverse / Kaggle, free for non-commercial research use.
Split defined	Yes	Scripted in the notebooks: 85/15, stratified by diagnosis, drawn from single-image lesions.
Hyperparameters	Yes	Adam, lr = 0.01, ϵ = 0.1; 150–300 epochs by backbone; early stopping; dropout 0.5.
Preprocessing	Yes	Resize to 299 ² (IRv2) or 224 ² ; augmentation to 8,000 images per class, applied after the split.
Random seeds	No	Not set. Run-to-run variance is unquantified; we will report mean $\pm$ std over repeated runs.
Trained weights	No	Not released; all results must be retrained from scratch.
Environment	Partial	TensorFlow 2.4.0 / Keras stated, but no <code>requirements.txt</code> and no container.
Compute reported	No	Training time and hardware are not stated.
Overall	4.5 / 7	— reproducible with identified gaps

We note this explicitly because it is a genuine methodological strength of the base paper.

Two weaknesses nonetheless remain, and they define the search space for our Milestone 3 hypothesis. First, restricting the test set to lesions photographed only once systematically *excludes* multi-image lesions from evaluation; such lesions are plausibly the larger or more clinically ambiguous ones, so the test set may be biased towards easier cases. Second, the 828-image test set contains on the order of ten samples each of `df` and `vasc`, so per-class metrics for the rare classes — and hence the reported 3.2% improvement — carry wide confidence intervals that the paper does not report. Both observations are consistent with the dataset audit of Abhishek *et al.* [12]. We emphasise that these are hypotheses to be tested in Milestones 3–4, not established conclusions.

VI. REPRODUCTION PLAN FOR MILESTONE 2

- 1) Obtain HAM10000 from the Harvard Dataverse and verify image count, class counts and `lesion_id` statistics against Table I.
- 2) Rebuild the environment. TensorFlow 2.4.0 is several years old; if it cannot be installed cleanly we will port the notebooks to a current TensorFlow release and document every change made.
- 3) Reproduce the *baseline* Inception-ResNet-v2 without attention and compare against the paper’s reported baseline.
- 4) Reproduce Inception-ResNet-v2 *with* Soft-Attention and compare against the target of 93.7% precision and 98.4% AUC.

- 5) Repeat both runs with three fixed seeds and report mean $\pm$ standard deviation, since the original work does not.
- 6) Report per-class sensitivity and precision, plus the confusion matrix. An exact match is not required; landing near the reported metric is.

VII. LIMITATIONS

This report is a literature survey and contains no experimental results of our own; all figures quoted are as reported by their original authors and have not yet been independently verified. The reported metrics in Table II are not mutually comparable, because the papers use different datasets, class counts, split protocols and augmentation strategies; the table is a summary, not a leaderboard. Our survey is restricted to English-language publications and to classification rather than segmentation, and it is not a systematic review in the PRISMA sense.

The dataset itself is limited. HAM10000 is drawn from two centres in Austria and Australia, and its patient population is predominantly fair-skinned; models trained on it should not be assumed to generalise across skin tones. The rare classes contain barely a hundred images each, so per-class estimates for `df` and `vasc` will be statistically fragile regardless of the method used.

Finally, and most importantly: **no clinical claims are made in this work**. Nothing in this project is validated for diagnostic use, none of the models discussed here are medical devices, and the reported metrics are benchmark results on a curated research dataset rather than evidence of clinical utility. Consistent with Tschandl *et al.* [7], the appropriate framing for this class of system is decision support for a qualified clinician, not autonomous diagnosis.

VIII. CONCLUSION

We surveyed ten representative papers on dermatoscopic skin-lesion classification, spanning clinical benchmarks, dataset descriptors, transfer-learning baselines, attention and imbalance methods, and dataset auditing. On the basis of six explicit criteria we selected Datta *et al.*’s Soft-Attention model as our base paper, principally because it is evaluated on HAM10000, releases public code for both its baseline and its proposed method, and reports sensitivity and AUC rather than accuracy alone. We completed a reproducibility checklist for it, scoring it 4.5/7 and naming the specific gaps — absent seeds, absent weights, and an unpinned environment — that we expect to encounter. We additionally verified its split logic directly from source, confirming that it avoids the lesion-level leakage common in this literature while identifying two residual weaknesses in its evaluation protocol that will motivate our Milestone 3 hypothesis. Milestone 2 will reproduce the baseline and the reported result.

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